CSA0404 – OPERATING SYSTEMS

EXPERIMENT 38

QUESTION

Design a C program to simulate SCAN disk scheduling algorithm.

PROGRAM

```c
                int temp = request_queue[i];
                request_queue[i] = request_queue[j];
                request_queue[j] = temp;
            }
        }
    }

    printf("\nSCAN (Elevator) Disk Scheduling:\n");

    int index = n;
    for (int i = 0; i < n; i++) {
        if (request_queue[i] >= head) {
            index = i;
            break;
        }
    }

    for (int i = index; i < n; i++) {
        seek_time += abs(head - request_queue[i]);
        head = request_queue[i];
    }

    for (int i = index - 1; i >= 0; i--) {
        seek_time += abs(head - request_queue[i]);
        head = request_queue[i];
    }

    printf("Total Seek Time: %d\n", seek_time);
    printf("Average Seek Time: %.2f\n", (float)seek_time / n);

    return 0;
}
```

```
Enter the number of disk requests: 7
Enter the disk request queue:
82 170 43 140 24 16 190
Enter the initial position of the disk head: 50

SCAN (Elevator) Disk Scheduling:
Total Seek Time: 314
Average Seek Time: 44.86

...Program finished with exit code 0
Press ENTER to exit console.
```